

UK Train Rides

Railway Transportation Analytics Dashboard

Project Code: CAI4_DAT2_S4

Team Members	
Habiba Mohammed Abdelfattah	Member
Gannah Salah	Member
Aya Hossam Mostafa	Member
Habiba Waleed Abdelrazek Mohammed	Project Leader
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GitHub: <https://github.com/Rodayna6/Train-Dash-Board>

April 2026

1. Project Planning & Management

1.1 Project Proposal

Overview

The UK Train Rides project is a data analytics and visualization initiative developed as part of the DEPI Round 4 Microsoft Power BI track. The project analyzes a real-world railway transportation dataset containing over 31,000 ticket transactions to generate actionable insights that support strategic decision-making for transport operators, authorities, and other key stakeholders.

Objectives

- Clean and preprocess raw railway ticketing data using Power BI's Power Query.
- Identify and formulate meaningful analytical questions answerable through the dataset.
- Build an interactive Power BI dashboard that visualizes ridership patterns, revenue performance, delay behavior, and ticket demand.
- Apply predictive analysis to forecast the number of rides and expected revenue for each day of the upcoming month.
- Produce a final report and presentation summarizing findings, methodology, and recommendations.

Scope

The project covers the full data analytics lifecycle: data ingestion, data cleaning, exploratory analysis, dashboard development, and predictive forecasting all conducted within the Microsoft Power BI ecosystem.

1.2 Project Plan — Timeline & Milestones

Week	Phase	Tasks	Deliverables
Week 1	Data Cleaning & Preprocessing	Import dataset into Power BI, handle missing values in Reason for Delay and Refund Request , normalize date/time columns, validate data types	Cleaned dataset ready for analysis; preprocessing documentation
Week 2	Analysis Questions Phase	Define key analytical questions for transport decision-makers; develop demand forecasting model; identify ticket class distribution trends	Set of analysis questions; predictive model outline.
Week 3	Dashboard Phase	Build interactive Power BI dashboard with filters, slicers, KPIs, and visual charts answering each analysis question	Published Power BI dashboard file
Week 4	Final Presentation	Prepare comprehensive report and presentation covering data analysis, model development, and key insights	Final report + presentation deck

Student Deadline: 4/20/2026 (Literature Review & Requirements) | 7/17/2026 (Final Presentation). All documents must be uploaded to GitHub.

1.3 Task Assignment & Roles

Team Member	Role	Responsibilities
Habiba Waleed Abdelrazek Mohammed	Project Leader	Overall coordination, milestone tracking, stakeholder communication, final review
Habiba Mohammed Abdelfattah	Data Analyst	Data cleaning, preprocessing in Power BI Power Query, data validation
Gannah Salah	Business Analyst	Formulating analysis questions, KPI definition, requirements documentation
Aya Hossam Mostafa	Dashboard Developer	Building Power BI visuals, designing dashboard layout and interactivity
Rodayna	Predictive Analyst	Building forecasting models for ride demand and revenue projection
Sara Mohamed	Documentation & Presentation	Writing reports, preparing presentation slides, GitHub management

1.4 Risk Assessment & Mitigation Plan

#	Risk	Likelihood	Impact	Mitigation
1	Missing/inconsistent data (e.g. Reason for Delay nulls)	High	Medium	Treat nulls as "No Delay"; document assumptions clearly in preprocessing notes
2	Incorrect date/time parsing across journey and purchase dates	Medium	High	Apply explicit date format conversion in Power Query; validate against known journey ranges
3	Team member unavailability near deadlines	Low	High	Distribute tasks across pairs; project leader maintains a shared status tracker on GitHub
4	Inaccurate demand forecasting due to limited historical data	Medium	Medium	Use Power BI's built-in forecasting with confidence intervals; clearly communicate forecast uncertainty
5	Dashboard performance issue with 31,000+ rows	Low	Medium	Aggregate data where possible; use data model optimization in Power BI
6	Scope creep from unplanned	Medium	Low	Freeze dashboard design at start of Week 3; change requests require team leader approval

1.5 KPIs (Key Performance Indicators)

The following KPIs define success for the UK Train Rides project:

Operational KPIs (Dashboard Quality)

- **Dashboard Completeness:** 100% of defined analysis questions answered by at least one visual.
- **Data Accuracy Rate:** $\geq 98\%$ of records correctly cleaned and classified post-preprocessing.
- **Filter Responsiveness:** Dashboard slicers and filters respond within 3 seconds on standard hardware.

Business KPIs — Insights Quality

- **On-Time Performance Rate:** Percentage of journeys with Journey Status = On Time (target: identify routes below 80%).
- **Revenue per Journey:** Average ticket Price segmented by Ticket Class and Ticket Type.
- **Refund Request Rate:** Proportion of delayed/cancelled journeys where Refund Request = Yes.
- **Demand Forecast Accuracy:** Predicted vs. actual rides for the forecasted month (target MAPE $<15\%$).
- **Railcard Utilization Rate:** Proportion of passengers using Railcard discounts vs. full-price tickets.

2. Literature Review

2.1 Feedback & Evaluation (Lecturer's Assessment Framework)

Data Preparation (20%): Quality of cleaning and preprocessing steps. Criteria include correct handling of null values (particularly in Reason for Delay and Refund Request), proper data type assignments, and thorough documentation of each transformation step in Power BI Power Query.

Analytical Depth (25%): Relevance and breadth of the analysis questions formulated. The instructor will assess whether the team identified non-obvious insights beyond simple counts — e.g., route-level delay patterns, seasonal demand trends, or payment method preferences.

Dashboard Design (25%): Visual clarity, logical layout, appropriate chart selection, and effective use of Power BI interactivity (slicers, drill-throughs, tooltips). The dashboard must clearly answer each stated analysis question.

Predictive Analysis (15%): Soundness of ride demand and revenue forecasting. The team should document assumptions, model selection rationale, and clearly communicate uncertainty through confidence intervals.

Documentation & Presentation (15%): Quality of the written report, structure of the final presentation, and completeness of GitHub uploads including all deliverables.

2.2 Suggested Improvements

Real-Time Data Integration: The current dataset is static. Integrating with a live railway API (e.g., National Rail's Darwin feed) would transform this into a live operational dashboard.

Geospatial Route Mapping: Adding a map visualization showing departure and arrival station coordinates would enable spatial analysis of high-delay corridors and high-revenue routes.

Passenger Segmentation: Deeper segmentation by Railcard type combined with Ticket Class and Purchase Type (online vs. station) could reveal behavioral differences between commuter and leisure traveler groups.

Delay Root Cause Analysis: A dedicated delay analysis page — including Pareto analysis of delay causes — would add significant operational value beyond current Reason for Delay breakdowns.

Advanced Forecasting: Replacing Power BI's built-in linear forecasting with ARIMA or exponential smoothing via Python integration would improve accuracy, especially for seasonal demand patterns.

Automated Refresh Pipeline: Setting up a Power BI scheduled refresh connected to a SharePoint or GitHub data source would prepare the solution for production deployment.

2.3 Final Grading Criteria

Component	Weight	Description
Project Documentation	20%	Completeness and professionalism of all documentation sections uploaded to GitHub
Data Cleaning & Preprocessing	15%	Accuracy, thoroughness, and documentation of all Power Query transformations
Analysis & Insights	25%	Depth, relevance, and correctness of analytical findings
Dashboard Implementation	25%	Design quality, interactivity, and completeness of the Power BI dashboard
Predictive Analysis	10%	Soundness of demand/revenue forecasting with documented assumptions
Final Presentation	5%	Clarity, structure, and team delivery during the final presentation session
Total	100%	

3. Requirements Gathering

3.1 Stakeholder Analysis

Stakeholder	Type	Needs & Expectations
Transport Operations Managers	Primary (End User)	Real-time visibility into journey status, delay patterns, and on-time performance by route
Revenue & Pricing Analysts	Primary (End User)	Insights into ticket class demand, revenue per route, discount utilization (Railcard, Advance, Off-Peak)
Customer Experience Teams	Secondary	Understanding refund request rates and delay reasons to improve service communications
DEPI Supervising Instructor	Evaluator	Complete, well-structured documentation; a functional, insightful dashboard; accurate predictive outputs
DEPI Program Administration	Regulatory	GitHub-hosted deliverables; adherence to project deadlines and DEPI R4 guidelines
Project Team	Developers	Clear task assignments, manageable scope, timely feedback, and a meaningful learning outcome

3.2 User Stories & Use Cases

User Story 1 — Operations Manager

"As an operations manager, I want to see which routes have the highest delay rates and the most common reasons for delay, so that I can prioritize infrastructure improvements."

Acceptance Criteria: Dashboard includes a route-level delay chart filtered by Departure Station and Arrival Destination, with a breakdown of Reason for Delay.

User Story 2 — Revenue Analyst

"As a revenue analyst, I want to compare ticket revenue across ticket types (Advance, Off-Peak, Anytime) and classes (Standard vs. First), so that I can optimize pricing strategy."

Acceptance Criteria: Dashboard displays a revenue breakdown matrix by Ticket Type and Ticket Class, filterable by date range and route.

User Story 3 — Demand Forecaster

"As a planning analyst, I want to see the projected number of rides and expected revenue for each day of the next month, so that I can allocate resources appropriately."

Acceptance Criteria: Dashboard includes a forecast chart with confidence intervals showing daily predicted rides and revenue for the upcoming month.

User Story 4 — Customer Experience Manager

"As a customer experience manager, I want to see what percentage of delayed or cancelled passengers requested refunds and which routes generated the most refund requests."

Acceptance Criteria: Dashboard shows Refund Request rate segmented by Journey Status, filterable by route and date.

User Story 5 — General Dashboard Consumer

"As any stakeholder, I want to filter all dashboard visuals by date range, departure station, arrival destination, and ticket class simultaneously."

Acceptance Criteria: All dashboard pages include synchronized slicers for date, station, and ticket class.

3.3 Functional Requirements

Requirement	Description
Data Import	Import the railway.csv dataset (31,653 rows, 18 columns) into Power BI.
Data Cleaning	The system shall handle null values in Reason for Delay (replacing with "No Delay") and Refund Request (replacing with "No").
Date/Time Parsing	The system shall correctly parse all date and time columns (Date of Purchase, Date of Journey, Departure Time, Arrival Time, Actual Arrival Time) as appropriate data types.
Delay Calculation	The system shall calculate actual delay duration (in minutes) by comparing Arrival Time to Actual Arrival Time.
Revenue Analysis	The system shall display total revenue, average ticket price, and revenue breakdown by Ticket Class, Ticket Type, and Railcard.
Journey Status Analysis	The system shall display the proportion of journeys that are On Time, Delayed, or Cancelled, filterable by route and date.
Route Analysis	The system shall identify the top and bottom performing routes by volume and revenue.
Demand Forecasting	The system shall forecast the daily number of rides for the next calendar month using Power BI's built-in analytics or equivalent.
Revenue Forecasting	The system shall forecast expected daily revenue for the next month, segmented by ticket class.
Interactive Filtering	The system shall provide synchronized slicers for: Date Range, Departure Station, Arrival Destination, Ticket Class, Purchase Type, and Payment Method.
Refund Analysis	The system shall display refund request rates segmented by journey status and route.
Export & Sharing	The final dashboard shall be publishable and shareable via Power BI Service or as a .pbix file on GitHub.

3.4 Non-Functional Requirements

Category	Requirement
Performance	Dashboard pages shall load and respond to filter interactions within 5 seconds on a standard laptop (8GB RAM).
Usability	All charts shall include clear titles, axis labels, and tooltips. Color schemes shall be consistent across all pages and accessible (color-blind friendly palette).
Reliability	The data model shall produce consistent results across all visuals — no conflicting figures between summary KPIs and detail charts.
Maintainability	The Power Query transformation steps shall be clearly named and documented so any team member can replicate or modify them.
Security	The dataset shall not be modified in its source form; all transformations shall occur within Power BI only. Raw data files shall be retained in their original state on GitHub.
Scalability	The data model design shall accommodate up to 200,000 rows without architectural changes, should the dataset be extended.
Accuracy	Forecasted values shall be presented with explicit confidence intervals. No forecast shall be presented without a stated assumption or model description.
Documentation	All transformation steps, analysis questions, and dashboard design decisions shall be documented in writing and uploaded to the project GitHub repository.